



Impact Influence

Configuration des équipements

1. Plan d'adressage IP :

a) Adressage IPv4 :

(Configuration du WAN) :

- Adresse sous-réseau WAN Bleu-Vert : 132.186.32.32 /29
- Adresse sous-réseau WAN Bleu-Rouge : 132.186.32.40 /29
- Adresse sous-réseau WAN Vert-Rouge : 132.186.32.48 /29

	Interface Se0/3/0 WAN 1	Interface Se0/3/1 WAN 2	Interface Fa0/0 LAN 1	Interface Fa0/1 LAN 2
Routeur bâtiment bleu	132.186.32.33 /29	132.186.32.41 /29	192.168.10.1 /24	192.168.20.1 /24
Routeur bâtiment vert	132.186.32.34 /29	132.186.32.49 /29	172.16.30.1 /29	172.16.40.1 /24
Routeur bâtiment rouge	132.186.32.42 /29	132.186.32.50 /29	10.0.0.1 /24	*

(Configuration du LAN) :

- Adresse LAN du bâtiment bleu : 192.168.0.0 /24
- Adresse LAN du bâtiment vert : 172.16.0.0 /24
- Adresse LAN du bâtiment rouge : 10.0.0.0 /24

Nom Vlan	ID Vlan	Interface	Adresse	CIDR	Masque de sous-réseau	Passerelle par défaut
Accueil	10	Fa0/0.10	192.168.10.0	/24	255.255.255.0	192.168.10.1
Showroom	20	Fa0/1.20	192.168.20.0	/24	255.255.255.0	192.168.20.1
Serveur Web	30	Fa0/0.30	172.16.30.0	/29	255.255.255.248	172.16.30.1
Réunion	40	Fa0/1.40	172.16.40.0	/24	255.255.255.0	172.16.40.1
Serveur DNS	50	Fa0/0.50	10.0.50.0	/29	255.255.255.248	10.0.50.1
Informatique	60	Fa0/0.60	10.0.60.0	/24	255.255.255.0	10.0.60.1
Finance	70	Fa0/0.70	10.0.70.0	/24	255.255.255.0	10.0.70.1
Commercial	80	Fa0/0.80	10.0.80.0	/24	255.255.255.0	10.0.80.1
RH	90	Fa0/0.90	10.0.90.0	/24	255.255.255.0	10.0.90.1
Direction	100	Fa0/0.100	10.0.100.0	/24	255.255.255.0	10.0.100.1

b) Adressage IPv6 :

(Configuration du WAN) :

- Adresse sous-réseau WAN Bleu-Vert : 2001:0:CAFE:0001::/127
- Adresse sous-réseau WAN Bleu-Rouge : 2001:0:CAFE:0002::/127
- Adresse sous-réseau WAN Vert-Rouge : 2001:0:CAFE:0003::/127

	Se0/3/0 WAN 1 global	Se0/3/0 WAN 1 link local	Se0/3/1 WAN 2 global	Se0/3/1 WAN 2 link local
Routeur bleu	2001:0:CAFE:0001::1	FE80::202:17FF:FEBD:6E01	2001:0:CAFE:0002::1	FE80::202:17FF:FEBD:6E01
Routeur vert	2001:0:CAFE:0001::2	FE80::204:9AFF:FE4D:B201	2001:0:CAFE:0003::1	FE80::204:9AFF:FE4D:B201
Routeur rouge	2001:0:CAFE:0002::2	FE80::2E0:8FFF:FE05:BD01	2001:0:CAFE:0003::2	FE80::2E0:8FFF:FE05:BD01

(Configuration du LAN) :

Légende : 2001:0:CAFE:X:YYYY

- 2001:0:CAFE est le préfixe global partagé
- X est un identifiant pour le bâtiment
- YYYY est un identifiant unique pour le VLAN dans ce bâtiment

Nom Vlan	ID Vlan	Sous-In- terface	Adresse globale	Adresse link local	Masque	Passerelle par défaut
Accueil	10	Fa 0/0.10	2001:0:CAFE:B1:A010::	FE80::202:17 FF:FEBD:6E0 1	/64	2001:0:CAFE:B1:A010 ::1
Showroom	20	Fa 0/1.20	2001:0:CAFE:B2:A020::	FE80::202:17 FF:FEBD:6E0 2	/64	2001:0:CAFE:B2:A020 ::1
Serveur Web	30	Fa 0/0.30	2001:0:CAFE:B3:A030::	FE80::204:9A FF:FE4D:B20 1	/64	2001:0:CAFE:B3:A030 ::1
Réunion	40	Fa 0/1.40	2001:0:CAFE:B4:A040::	FE80::204:9A FF:FE4D:B20 2	/64	2001:0:CAFE:B4:A040 ::1
Serveur DNS	50	Fa 0/0.50	2001:0:CAFE:B5:A050::	FE80::2E0:8F FF:FE05:BD0 1	/64	2001:0:CAFE:B5:A050 ::1
Informatique	60	Fa 0/0.60	2001:0:CAFE:B6:A060::	FE80::2E0:8F FF:FE05:BD0 1	/64	2001:0:CAFE:B6:A060 ::1
Finance	70	Fa 0/0.70	2001:0:CAFE:B7:A070::	FE80::2E0:8F FF:FE05:BD0 1	/64	2001:0:CAFE:B7:A070 ::1
Commercial	80	Fa 0/0.80	2001:0:CAFE:B8:A080::	FE80::2E0:8F FF:FE05:BD0 1	/64	2001:0:CAFE:B8:A080 ::1
RH	90	Fa 0/0.90	2001:0:CAFE:B9:A090::	FE80::2E0:8F FF:FE05:BD0 1	/64	2001:0:CAFE:B9:A090 ::1
Direction	100	Fa 0/0.100	2001:0:CAFE:BA:A0A0::	FE80::2E0:8F FF:FE05:BD0 1	/64	2001:0:CAFE:BA:A0A0 ::1

2. Configuration DNS :

Server_DNS

PhysicalConfig**Services**DesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record**

Address

Add

Save

Remove

No.	Name	Type	Detail
0	impactinfluence.com	A Record	132.186.32.49
1	impactinfluencev6.com	AAAA Record	2001:0:CAFE:B3:A030::2

DNS Cache

☐ Top

Configuration DNS (suite) :

Server_DNS

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

10.0.50.6

Subnet Mask

255.255.255.0

Default Gateway

10.0.50.1

DNS Server

10.0.50.6

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

2001:0:CAFE:B5:A0F0::1

/ 64

Link Local Address

FE80::20D:BDFF:FE1D:D593

Default Gateway

2001:0:CAFE:B5:A050::1

DNS Server

2001:0:CAFE:B5:A0F0::1

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

3. Configuration des routeurs (comprenant les tables de routage, les sous-interfaces et les NAT) :

a) Routeur Rouge :

(Show ip route) :

```
Routeur_Rouge#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.50.0/24 is directly connected, FastEthernet0/0.50
L       10.0.50.1/32 is directly connected, FastEthernet0/0.50
C       10.0.60.0/24 is directly connected, FastEthernet0/0.60
L       10.0.60.1/32 is directly connected, FastEthernet0/0.60
C       10.0.70.0/24 is directly connected, FastEthernet0/0.70
L       10.0.70.1/32 is directly connected, FastEthernet0/0.70
C       10.0.80.0/24 is directly connected, FastEthernet0/0.80
L       10.0.80.1/32 is directly connected, FastEthernet0/0.80
C       10.0.90.0/24 is directly connected, FastEthernet0/0.90
L       10.0.90.1/32 is directly connected, FastEthernet0/0.90
C       10.0.100.0/24 is directly connected, FastEthernet0/0.100
L       10.0.100.1/32 is directly connected, FastEthernet0/0.100
O       132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
O       132.186.32.32/29 [110/128] via 132.186.32.41, 00:06:42, Serial0/3/0
        [110/128] via 132.186.32.49, 00:06:42, Serial0/3/1
C       132.186.32.40/29 is directly connected, Serial0/3/0
L       132.186.32.42/32 is directly connected, Serial0/3/0
C       132.186.32.48/29 is directly connected, Serial0/3/1
L       132.186.32.50/32 is directly connected, Serial0/3/1
O       172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
O       172.16.30.0/29 [110/65] via 132.186.32.49, 00:06:42, Serial0/3/1
O       172.16.40.0/24 [110/65] via 132.186.32.49, 00:06:42, Serial0/3/1
O       192.168.10.0/24 [110/65] via 132.186.32.41, 00:06:42, Serial0/3/0
O       192.168.20.0/24 [110/65] via 132.186.32.41, 00:06:42, Serial0/3/0
```

(Show ipv6 route) :

```
Routeur_Rouge#show ipv6 route
IPv6 Routing Table - 23 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
O 2001:0:CAFE:1::/127 [110/128]
    via FE80::202:17FF:FEBD:6E01, Serial0/3/0
O 2001:0:CAFE:1::2/127 [110/128]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/1
O 2001:0:CAFE:2::/127 [110/128]
    via FE80::202:17FF:FEBD:6E01, Serial0/3/0
C 2001:0:CAFE:2::2/127 [0/0]
    via Serial0/3/0, directly connected
L 2001:0:CAFE:2::2/128 [0/0]
    via Serial0/3/0, receive
O 2001:0:CAFE:3::/127 [110/128]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/1
C 2001:0:CAFE:3::2/127 [0/0]
    via Serial0/3/1, directly connected
L 2001:0:CAFE:3::2/128 [0/0]
    via Serial0/3/1, receive
O 2001:0:CAFE:B3::/64 [110/65]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/1
O 2001:0:CAFE:B4::/64 [110/65]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/1
C 2001:0:CAFE:B5::/64 [0/0]
    via FastEthernet0/0.50, directly connected
L 2001:0:CAFE:B5:A050::1/128 [0/0]
    via FastEthernet0/0.50, receive
C 2001:0:CAFE:B6::/64 [0/0]
    via FastEthernet0/0.60, directly connected
L 2001:0:CAFE:B6:A060::1/128 [0/0]
    via FastEthernet0/0.60, receive
C 2001:0:CAFE:B7::/64 [0/0]
    via FastEthernet0/0.70, directly connected
L 2001:0:CAFE:B7:A070::1/128 [0/0]
    via FastEthernet0/0.70, receive
```

```
C 2001:0:CAFE:B8::/64 [0/0]
    via FastEthernet0/0.80, directly connected
L 2001:0:CAFE:B8:A080::1/128 [0/0]
    via FastEthernet0/0.80, receive
C 2001:0:CAFE:B9::/64 [0/0]
    via FastEthernet0/0.90, directly connected
L 2001:0:CAFE:B9:A090::1/128 [0/0]
    via FastEthernet0/0.90, receive
C 2001:0:CAFE:BA::/64 [0/0]
    via FastEthernet0/0.100, directly connected
L 2001:0:CAFE:BA:A0A0::1/128 [0/0]
    via FastEthernet0/0.100, receive
L FF00::/8 [0/0]
    via Null0, receive
```

(Show ip interface brief) :

```

Routeur_Rouge#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES manual up           up
FastEthernet0/0.50       10.0.50.1       YES manual up           up
FastEthernet0/0.60       10.0.60.1       YES manual up           up
FastEthernet0/0.70       10.0.70.1       YES manual up           up
FastEthernet0/0.80       10.0.80.1       YES manual up           up
FastEthernet0/0.90       10.0.90.1       YES manual up           up
FastEthernet0/0.100      10.0.100.1      YES manual up           up
FastEthernet0/1          unassigned      YES unset administratively down down
Serial0/3/0              132.186.32.42   YES manual up           up
Serial0/3/1              132.186.32.50   YES manual up           up
Vlan1                    unassigned      YES unset administratively down down
Vlan50                   unassigned      YES unset down           down
Vlan60                   unassigned      YES unset down           down
Vlan70                   unassigned      YES unset down           down
Vlan80                   unassigned      YES unset down           down
Vlan90                   unassigned      YES unset down           down
Vlan100                  unassigned      YES unset down           down

```

(Show ipv6 interface brief) :

```

Routeur_Rouge#show ipv6 interface brief
FastEthernet0/0          [up/up]
unassigned
FastEthernet0/0.50       [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:B5:A050::1
FastEthernet0/0.60       [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:B6:A060::1
FastEthernet0/0.70       [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:B7:A070::1
FastEthernet0/0.80       [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:B8:A080::1
FastEthernet0/0.90       [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:B9:A090::1
FastEthernet0/0.100      [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:BA:A0A0::1
FastEthernet0/1          [administratively down/down]
unassigned
Serial0/3/0              [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:2::2
Serial0/3/1              [up/up]
FE80::2E0:8FFF:FE05:BD01
2001:0:CAFE:3::2
Vlan1                    [administratively down/down]
unassigned
Vlan50                   [down/down]
unassigned
Vlan60                   [down/down]
unassigned
Vlan70                   [down/down]
unassigned
-----
Vlan80                   [down/down]
unassigned
Vlan90                   [down/down]
unassigned
Vlan100                  [down/down]
unassigned

```


(NAT Overload) :

```
ip nat inside source list 1 interface Serial0/3/1 overload
ip nat inside source list 2 interface Serial0/3/1 overload
ip nat inside source list 3 interface Serial0/3/1 overload
ip nat inside source list 4 interface Serial0/3/1 overload
ip nat inside source list 5 interface Serial0/3/1 overload
ip nat inside source list 6 interface Serial0/3/1 overload
ip classless
!
ip flow-export version 9
!
!
access-list 1 permit 10.0.50.0 0.0.0.255
access-list 2 permit 10.0.60.0 0.0.0.255
access-list 3 permit 10.0.70.0 0.0.0.255
access-list 4 permit 10.0.80.0 0.0.0.255
access-list 5 permit 10.0.90.0 0.0.0.255
access-list 6 permit 10.0.100.0 0.0.0.255
```

(ACL VLAN IPv4) :

```
ip access-list extended VLAN_ACL
permit icmp 10.0.60.0 0.0.0.255 172.16.30.0 0.0.0.7 echo
permit icmp 10.0.60.0 0.0.0.255 10.0.50.0 0.0.0.255 echo
permit icmp 10.0.60.0 0.0.0.255 10.0.70.0 0.0.0.255 echo
permit icmp 10.0.60.0 0.0.0.255 10.0.80.0 0.0.0.255 echo
permit icmp 10.0.60.0 0.0.0.255 10.0.90.0 0.0.0.255 echo
permit icmp 10.0.60.0 0.0.0.255 10.0.100.0 0.0.0.255 echo
permit udp 10.0.60.0 0.0.0.255 10.0.50.0 0.0.0.255 eq domain
permit tcp 10.0.60.0 0.0.0.255 any eq www
permit tcp 10.0.60.0 0.0.0.255 any eq 443
permit tcp 10.0.60.0 0.0.0.255 172.16.30.0 0.0.0.7 established
permit tcp 10.0.60.0 0.0.0.255 10.0.70.0 0.0.0.255 established
permit tcp 10.0.60.0 0.0.0.255 10.0.80.0 0.0.0.255 established
permit tcp 10.0.60.0 0.0.0.255 10.0.90.0 0.0.0.255 established
permit tcp 10.0.60.0 0.0.0.255 10.0.100.0 0.0.0.255 established
deny ip any any
```

(ACL VLAN IPv6) :

```
ipv6 access-list VLAN_ACL_V6
permit icmp 2001:0:CAFE:B6::/64 2001:0:CAFE:B3::/64 echo-request
permit icmp 2001:0:CAFE:B6::/64 2001:0:CAFE:B5::/64 echo-request
permit icmp 2001:0:CAFE:B6::/64 2001:0:CAFE:B7::/64 echo-request
permit icmp 2001:0:CAFE:B6::/64 2001:0:CAFE:B8::/64 echo-request
permit icmp 2001:0:CAFE:B6::/64 2001:0:CAFE:B9::/64 echo-request
permit icmp 2001:0:CAFE:B6::/64 2001:0:CAFE:BA::/64 echo-request
permit udp 2001:0:CAFE:B6::/64 2001:0:CAFE:B5::/64 eq domain
permit tcp 2001:0:CAFE:B6::/64 any eq www
permit tcp 2001:0:CAFE:B6::/64 any eq 443
permit tcp 2001:0:CAFE:B6::/64 2001:0:CAFE:B7::/64
permit tcp 2001:0:CAFE:B6::/64 2001:0:CAFE:B8::/64
permit tcp 2001:0:CAFE:B6::/64 2001:0:CAFE:B9::/64
permit tcp 2001:0:CAFE:B6::/64 2001:0:CAFE:BA::/64
deny ipv6 any any
```

(ACL serveur DNS) :

```
ip access-list extended DNS_ACL
permit icmp 10.0.60.0 0.0.0.255 host 10.0.50.6 echo
permit icmp 10.0.70.0 0.0.0.255 host 10.0.50.6 echo
permit icmp 10.0.80.0 0.0.0.255 host 10.0.50.6 echo
permit icmp 10.0.90.0 0.0.0.255 host 10.0.50.6 echo
permit icmp 10.0.100.0 0.0.0.255 host 10.0.50.6 echo
permit udp 10.0.60.0 0.0.0.255 host 10.0.50.6 eq domain
permit udp 10.0.70.0 0.0.0.255 host 10.0.50.6 eq domain
permit udp 10.0.80.0 0.0.0.255 host 10.0.50.6 eq domain
permit udp 10.0.90.0 0.0.0.255 host 10.0.50.6 eq domain
permit udp 10.0.100.0 0.0.0.255 host 10.0.50.6 eq domain
permit tcp 10.0.60.0 0.0.0.255 host 10.0.50.6 established
permit tcp 10.0.70.0 0.0.0.255 host 10.0.50.6 established
permit tcp 10.0.80.0 0.0.0.255 host 10.0.50.6 established
permit tcp 10.0.90.0 0.0.0.255 host 10.0.50.6 established
permit tcp 10.0.100.0 0.0.0.255 host 10.0.50.6 established
```

(Show running-config) :

```
Routeur_Rouge#show run
Building configuration...

Current configuration : 6553 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Routeur_Rouge
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
ipv6 dhcp pool DNSv6
 dns-server 2001::CAFE:B5:A0F0::1
!
!
!
!
license udi pid CISCO2811/K9 sn FTX1017R8IO-
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
```

```

interface FastEthernet0/0
no ip address
ip nat inside
duplex auto
speed auto
!
interface FastEthernet0/0.50
encapsulation dot1Q 50
ip address 10.0.50.1 255.255.255.0
ip ospf 1 area 0
ip access-group DNS_ACL out
ip nat inside
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B5:A050::1/64
ipv6 nd other-config-flag
ipv6 ospf 2 area 0
!
interface FastEthernet0/0.60
encapsulation dot1Q 60
ip address 10.0.60.1 255.255.255.0
ip ospf 1 area 0
ip access-group VLAN_ACL in
ipv6 traffic-filter VLAN_ACL_V6 in
ip nat inside
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B6:A060::1/64
ipv6 nd other-config-flag
ipv6 ospf 2 area 0
ipv6 dhcp server DNSv6
!
interface FastEthernet0/0.70
encapsulation dot1Q 70
ip address 10.0.70.1 255.255.255.0
ip ospf 1 area 0
ip nat inside
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B7:A070::1/64
ipv6 nd other-config-flag
ipv6 ospf 2 area 0
ipv6 dhcp server DNSv6

interface FastEthernet0/0.80
encapsulation dot1Q 80
ip address 10.0.80.1 255.255.255.0
ip ospf 1 area 0
ip nat inside
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B8:A080::1/64
ipv6 nd other-config-flag
ipv6 ospf 2 area 0
ipv6 dhcp server DNSv6
!
interface FastEthernet0/0.90
encapsulation dot1Q 90
ip address 10.0.90.1 255.255.255.0
ip ospf 1 area 0
ip nat inside
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B9:A090::1/64
ipv6 nd other-config-flag
ipv6 ospf 2 area 0
ipv6 dhcp server DNSv6
!
interface FastEthernet0/0.100
encapsulation dot1Q 100
ip address 10.0.100.1 255.255.255.0
ip ospf 1 area 0
ip nat inside
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:BA:A0A0::1/64
ipv6 nd other-config-flag
ipv6 ospf 2 area 0
ipv6 dhcp server DNSv6
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown

```

```

interface Serial0/3/0
ip address 132.186.32.42 255.255.255.248
ip nat outside
ipv6 address 2001:0:CAFE:2::2/127
ipv6 ospf 2 area 0
!
interface Serial0/3/1
ip address 132.186.32.50 255.255.255.248
ip nat outside
ipv6 address 2001:0:CAFE:3::2/127
ipv6 ospf 2 area 0
clock rate 2000000
,

```

```

router ospf 1
log-adjacency-changes
network 132.186.32.40 0.0.0.7 area 0
network 132.186.32.48 0.0.0.7 area 0
network 10.0.0.0 0.0.0.255 area 0
network 10.0.50.0 0.0.0.7 area 0
network 10.0.60.0 0.0.0.255 area 0
network 10.0.70.0 0.0.0.255 area 0
network 10.0.80.0 0.0.0.255 area 0
network 10.0.90.0 0.0.0.255 area 0
network 10.0.100.0 0.0.0.255 area 0
!
ipv6 router ospf 2
router-id 3.3.3.3
log-adjacency-changes

```

b) Routeur Vert :

(Show ip route) :

```

Routeur Vert# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 6 subnets
O       10.0.50.0/24 [110/65] via 132.186.32.50, 00:15:22, Serial0/3/1
O       10.0.60.0/24 [110/65] via 132.186.32.50, 00:15:22, Serial0/3/1
O       10.0.70.0/24 [110/65] via 132.186.32.50, 00:15:22, Serial0/3/1
O       10.0.80.0/24 [110/65] via 132.186.32.50, 00:15:22, Serial0/3/1
O       10.0.90.0/24 [110/65] via 132.186.32.50, 00:15:22, Serial0/3/1
O       10.0.100.0/24 [110/65] via 132.186.32.50, 00:15:22, Serial0/3/1
    132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       132.186.32.32/29 is directly connected, Serial0/3/0
L       132.186.32.34/32 is directly connected, Serial0/3/0
O       132.186.32.40/29 [110/128] via 132.186.32.50, 00:15:22, Serial0/3/1
        [110/128] via 132.186.32.33, 00:15:22, Serial0/3/0
C       132.186.32.48/29 is directly connected, Serial0/3/1
L       132.186.32.49/32 is directly connected, Serial0/3/1
    172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.16.30.0/29 is directly connected, FastEthernet0/0.30
L       172.16.30.1/32 is directly connected, FastEthernet0/0.30
C       172.16.40.0/24 is directly connected, FastEthernet0/1.40
L       172.16.40.1/32 is directly connected, FastEthernet0/1.40
O       192.168.10.0/24 [110/65] via 132.186.32.33, 00:15:22, Serial0/3/0
O       192.168.20.0/24 [110/65] via 132.186.32.33, 00:15:22, Serial0/3/0

```

(Show ipv6 route) :

```
Routeur_Vert# show ipv6 route
IPv6 Routing Table - 19 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
O  2001:0:CAFE:1::/127 [110/128]
    via FE80::202:17FF:FEBD:6E01, Serial0/3/0
C  2001:0:CAFE:1::2/127 [0/0]
    via Serial0/3/0, directly connected
L  2001:0:CAFE:1::2/128 [0/0]
    via Serial0/3/0, receive
O  2001:0:CAFE:2::/127 [110/128]
    via FE80::202:17FF:FEBD:6E01, Serial0/3/0
O  2001:0:CAFE:2::2/127 [110/128]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
C  2001:0:CAFE:3::/127 [0/0]
    via Serial0/3/1, directly connected
L  2001:0:CAFE:3::1/128 [0/0]
    via Serial0/3/1, receive
O  2001:0:CAFE:3::2/127 [110/128]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
C  2001:0:CAFE:B3::/64 [0/0]
    via FastEthernet0/0.30, directly connected
L  2001:0:CAFE:B3:A030::1/128 [0/0]
    via FastEthernet0/0.30, receive
C  2001:0:CAFE:B4::/64 [0/0]
    via FastEthernet0/1.40, directly connected
L  2001:0:CAFE:B4:A040::1/128 [0/0]
    via FastEthernet0/1.40, receive
O  2001:0:CAFE:B5::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:B6::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:B7::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:B8::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1

O  2001:0:CAFE:B9::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:BA::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
L  FF00::/8 [0/0]
    via Null0, receive
```

(Show running-config) :

```
Routeur_Vert#show run
Building configuration...

Current configuration : 1667 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Routeur_Vert
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
license udi pid CISCO2811/K9 sn FTX1017F16N-
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
```

```

interface FastEthernet0/0
no ip address
ip ospf 1 area 0
ip nat inside
duplex auto
speed auto
!
interface FastEthernet0/0.30
encapsulation dot1Q 30
ip address 172.16.30.1 255.255.255.248
ip nat inside
ipv6 address 2001:0:CAFE:B3:A030::1/64
ipv6 ospf 2 area 0
!
interface FastEthernet0/1
no ip address
ip ospf 1 area 0
duplex auto
speed auto
!
interface FastEthernet0/1.40
encapsulation dot1Q 40
ip address 172.16.40.1 255.255.255.0
ipv6 address 2001:0:CAFE:B4:A040::1/64
ipv6 ospf 2 area 0
!
interface Serial0/3/0
ip address 132.186.32.34 255.255.255.248
ipv6 address 2001:0:CAFE:1::2/127
ipv6 ospf 2 area 0
!
interface Serial0/3/1
ip address 132.186.32.49 255.255.255.248
ip nat outside
ipv6 address 2001:0:CAFE:3::1/127
ipv6 ospf 2 area 0
!
interface Vlan1
no ip address
shutdown
,

router ospf 1
log-adjacency-changes
network 132.186.32.32 0.0.0.7 area 0
network 132.186.32.48 0.0.0.7 area 0
network 172.16.0.0 0.0.0.255 area 0
network 172.16.30.0 0.0.0.7 area 0
network 172.16.40.0 0.0.0.255 area 0
!
ipv6 router ospf 2
router-id 2.2.2.2
log-adjacency-changes
!
ip nat inside source static tcp 172.16.30.2 80 132.186.32.49 80
ip nat inside source static tcp 172.16.30.2 443 132.186.32.49 443
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end

```

c) Routeur Bleu :

(Show ip route) :

```
Routeur_Bleu# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 6 subnets
O       10.0.50.0/24 [110/65] via 132.186.32.42, 00:32:04, Serial0/3/1
O       10.0.60.0/24 [110/65] via 132.186.32.42, 00:32:04, Serial0/3/1
O       10.0.70.0/24 [110/65] via 132.186.32.42, 00:32:04, Serial0/3/1
O       10.0.80.0/24 [110/65] via 132.186.32.42, 00:32:04, Serial0/3/1
O       10.0.90.0/24 [110/65] via 132.186.32.42, 00:32:04, Serial0/3/1
O       10.0.100.0/24 [110/65] via 132.186.32.42, 00:32:04, Serial0/3/1
    132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       132.186.32.32/29 is directly connected, Serial0/3/0
L       132.186.32.33/32 is directly connected, Serial0/3/0
C       132.186.32.40/29 is directly connected, Serial0/3/1
L       132.186.32.41/32 is directly connected, Serial0/3/1
O       132.186.32.48/29 [110/128] via 132.186.32.34, 00:32:04, Serial0/3/0
        [110/128] via 132.186.32.42, 00:32:04, Serial0/3/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
O       172.16.30.0/29 [110/65] via 132.186.32.34, 00:32:14, Serial0/3/0
O       172.16.40.0/24 [110/65] via 132.186.32.34, 00:32:14, Serial0/3/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/0.10
L       192.168.10.1/32 is directly connected, FastEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, FastEthernet0/1.20
L       192.168.20.1/32 is directly connected, FastEthernet0/1.20
```


(show ipv6 route) :

```
Routeur Bleu# show ipv6 route
IPv6 Routing Table - 21 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
C  2001:0:CAFE:1::/127 [0/0]
    via Serial0/3/0, directly connected
L  2001:0:CAFE:1::1/128 [0/0]
    via Serial0/3/0, receive
O  2001:0:CAFE:1::2/127 [110/128]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/0
C  2001:0:CAFE:2::/127 [0/0]
    via Serial0/3/1, directly connected
L  2001:0:CAFE:2::1/128 [0/0]
    via Serial0/3/1, receive
O  2001:0:CAFE:2::2/127 [110/128]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:3::/127 [110/128]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/0
O  2001:0:CAFE:3::2/127 [110/128]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
C  2001:0:CAFE:B1::/64 [0/0]
    via FastEthernet0/0.10, directly connected
L  2001:0:CAFE:B1:A010::1/128 [0/0]
    via FastEthernet0/0.10, receive
C  2001:0:CAFE:B2::/64 [0/0]
    via FastEthernet0/1.20, directly connected
L  2001:0:CAFE:B2:A020::1/128 [0/0]
    via FastEthernet0/1.20, receive
O  2001:0:CAFE:B3::/64 [110/65]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/0
O  2001:0:CAFE:B4::/64 [110/65]
    via FE80::204:9AFF:FE4D:B201, Serial0/3/0
O  2001:0:CAFE:B5::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:B6::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1

O  2001:0:CAFE:B7::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:B8::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:B9::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
O  2001:0:CAFE:BA::/64 [110/65]
    via FE80::2E0:8FFF:FE05:BD01, Serial0/3/1
L  FF00::/8 [0/0]
    via Null0, receive
```

(Show running-config) :

```
Routeur_Bleu#show run
Building configuration...

Current configuration : 1600 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Routeur_Bleu
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
license udi pid CISCO2811/K9 sn FTX1017ZL8G-
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
```

```

interface FastEthernet0/0
no ip address
ip ospf 1 area 0
duplex auto
speed auto
!
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
ip ospf 1 area 0
ipv6 address 2001:0:CAFE:B1:A010::1/64
!
interface FastEthernet0/0.20
no ip address
!
interface FastEthernet0/1
no ip address
ip ospf 1 area 0
duplex auto
speed auto
!
interface FastEthernet0/1.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip ospf 1 area 0
ipv6 address 2001:0:CAFE:B2:A020::1/64
!
interface Serial0/3/0
ip address 132.186.32.33 255.255.255.248
ipv6 address 2001:0:CAFE:1::1/127
ipv6 ospf 2 area 0
clock rate 2000000
!
interface Serial0/3/1
ip address 132.186.32.41 255.255.255.248
ipv6 address 2001:0:CAFE:2::1/127
ipv6 ospf 2 area 0
clock rate 2000000

```

```

interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 132.186.32.40 0.0.0.7 area 0
network 132.186.32.32 0.0.0.7 area 0
network 192.168.0.0 0.0.0.255 area 0
!
ipv6 router ospf 2
router-id 1.1.1.1
log-adjacency-changes
!
ip nat inside source list 1 interface Serial0/3/1 overload
ip classless
!
ip flow-export version 9
!
!
access-list 1 permit 0.0.0.0 255.255.255.0
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end

```

4. Configuration des switches layer 3 (comprenant les paramètres DHCP, les VLAN, le LACP) :

a) Switch Core :

(configuration trunk) :

```
Switch_Core# show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/3     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     50,60,70,80,90,100
Fa0/2     50,60,70,80,90,100
Fa0/3     50,60,70,80,90,100

Port      Vlans allowed and active in management domain
Fa0/1     50,60,70,80,90,100
Fa0/2     50,60,70,80,90,100
Fa0/3     50,60,70,80,90,100

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     50,60,70,80,90,100
Fa0/2     none
Fa0/3     50,60,70,80,90,100

Switch_Core#
```

b) Switch Distributed 1 :

(show running-config / DHCP) :

```
Switch#show run
Building configuration...

Current configuration : 4963 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
ip dhcp excluded-address 10.0.50.1
ip dhcp excluded-address 10.0.50.2
ip dhcp excluded-address 10.0.50.3
ip dhcp excluded-address 10.0.60.1
ip dhcp excluded-address 10.0.60.2
ip dhcp excluded-address 10.0.60.3
ip dhcp excluded-address 10.0.70.1
ip dhcp excluded-address 10.0.70.2
ip dhcp excluded-address 10.0.70.3
ip dhcp excluded-address 10.0.80.1
ip dhcp excluded-address 10.0.80.2
ip dhcp excluded-address 10.0.80.3
ip dhcp excluded-address 10.0.90.1
ip dhcp excluded-address 10.0.90.2
ip dhcp excluded-address 10.0.90.3
ip dhcp excluded-address 10.0.100.1
ip dhcp excluded-address 10.0.100.2
ip dhcp excluded-address 10.0.100.3
ip dhcp excluded-address 10.0.50.6
```

```

ip dhcp pool vlan50
network 10.0.50.0 255.255.255.248
default-router 10.0.50.1
dns-server 10.0.50.6
ip dhcp pool vlan60
network 10.0.60.0 255.255.255.0
default-router 10.0.60.1
dns-server 10.0.50.6
ip dhcp pool vlan70
network 10.0.70.0 255.255.255.0
default-router 10.0.70.1
dns-server 10.0.50.6
ip dhcp pool vlan80
network 10.0.80.0 255.255.255.0
default-router 10.0.80.1
dns-server 10.0.50.6
ip dhcp pool vlan90
network 10.0.90.0 255.255.255.0
default-router 10.0.90.1
dns-server 10.0.50.6
ip dhcp pool vlan100
network 10.0.100.0 255.255.255.0
dns-server 10.0.50.6
!

```

```

.
spanning-tree mode pvst
!
!
!
!
!
!
interface Port-channel1
switchport trunk allowed vlan 50,60,70,80,90,100
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk allowed vlan 50,60,70,80,90,100
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk allowed vlan 50,60,70,80,90,100
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport trunk allowed vlan 50,60,70,80,90,100
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/4
switchport trunk allowed vlan 50,60,70,80,90,100
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/5
switchport trunk allowed vlan 50,60,70,80,90,100
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode active
.

```

```

interface FastEthernet0/6
  switchport trunk allowed vlan 50
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface FastEthernet0/7
  switchport trunk allowed vlan 60
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface FastEthernet0/8
  switchport trunk allowed vlan 70
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface FastEthernet0/9
  switchport trunk allowed vlan 80
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface FastEthernet0/10
  switchport trunk allowed vlan 90
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface FastEthernet0/11
  switchport trunk allowed vlan 100
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!

```

```

interface Vlan1
  no ip address
  shutdown
!
interface Vlan50
  mac-address 00d0.ff0c.6e01
  ip address 10.0.50.2 255.255.255.0
  ip helper-address 10.0.50.3
  ipv6 nd ra suppress
  ipv6 address 2001:0:CAFE:B5:A050::2/64
  ipv6 nd other-config-flag
!
interface Vlan60
  mac-address 00d0.ff0c.6e02
  ip address 10.0.60.2 255.255.255.0
  ip helper-address 10.0.60.3
  ipv6 nd ra suppress
  ipv6 address 2001:0:CAFE:B6:A060::2/64
  ipv6 nd other-config-flag
!
interface Vlan70
  mac-address 00d0.ff0c.6e03
  ip address 10.0.70.2 255.255.255.0
  ip helper-address 10.0.70.3
  ipv6 nd ra suppress
  ipv6 address 2001:0:CAFE:B7:A070::2/64
  ipv6 nd other-config-flag
!
interface Vlan80
  mac-address 00d0.ff0c.6e04
  ip address 10.0.80.2 255.255.255.0
  ip helper-address 10.0.80.3
  ipv6 nd ra suppress
  ipv6 address 2001:0:CAFE:B8:A080::2/64
  ipv6 nd other-config-flag

```

```

interface Vlan90
  mac-address 00d0.ff0c.6e05
  ip address 10.0.90.2 255.255.255.0
  ip helper-address 10.0.90.3
  ipv6 nd ra suppress
  ipv6 address 2001:0:CAFE:B9:A090::2/64
  ipv6 nd other-config-flag
!
interface Vlan100
  mac-address 00d0.ff0c.6e06
  ip address 10.0.100.2 255.255.255.0
  ip helper-address 10.0.100.3
  ipv6 nd ra suppress
  ipv6 address 2001:0:CAFE:BA:A0A0::2/64
  ipv6 nd other-config-flag
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
!
end

```

(show etherchannel summary):

```

Switch#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SU)        LACP       Fa0/2(P) Fa0/3(P) Fa0/4(P) Fa0/5(P)

```

c) Switch Distributed 2 :

(show running-config / DHCP) :

```
Switch#show run
Building configuration...

Current configuration : 4849 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
ip dhcp excluded-address 10.0.50.1
ip dhcp excluded-address 10.0.50.2
ip dhcp excluded-address 10.0.50.3
ip dhcp excluded-address 10.0.50.4
ip dhcp excluded-address 10.0.60.1
ip dhcp excluded-address 10.0.60.2
ip dhcp excluded-address 10.0.60.3
ip dhcp excluded-address 10.0.70.1
ip dhcp excluded-address 10.0.70.2
ip dhcp excluded-address 10.0.70.3
ip dhcp excluded-address 10.0.80.1
ip dhcp excluded-address 10.0.80.2
ip dhcp excluded-address 10.0.80.3
ip dhcp excluded-address 10.0.90.1
ip dhcp excluded-address 10.0.90.2
ip dhcp excluded-address 10.0.90.3
ip dhcp excluded-address 10.0.100.1
ip dhcp excluded-address 10.0.100.2
ip dhcp excluded-address 10.0.100.3
ip dhcp excluded-address 10.0.50.6
```

```
ip dhcp pool vlan50
 network 10.0.50.0 255.255.255.248
 default-router 10.0.50.1
 dns-server 10.0.50.6
ip dhcp pool vlan60
 network 10.0.60.0 255.255.255.0
 default-router 10.0.60.1
 dns-server 10.0.50.6
ip dhcp pool vlan70
 network 10.0.70.0 255.255.255.0
 default-router 10.0.70.1
 dns-server 10.0.50.6
ip dhcp pool vlan80
 network 10.0.80.0 255.255.255.0
 default-router 10.0.80.1
 dns-server 10.0.50.6
ip dhcp pool vlan90
 network 10.0.90.0 255.255.255.0
 default-router 10.0.90.1
 dns-server 10.0.50.6
ip dhcp pool vlan100
 network 10.0.100.0 255.255.255.0
 default-router 10.0.100.1
 dns-server 10.0.50.6
```



```

spanning-tree mode pvst
!
!
!
!
!
!
interface Port-channel1
 switchport trunk allowed vlan 50,60,70,80,90,100
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/1
 switchport trunk allowed vlan 50,60,70,80,90,100
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 50,60,70,80,90,100
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface FastEthernet0/3
 switchport trunk allowed vlan 50,60,70,80,90,100
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface FastEthernet0/4
 switchport trunk allowed vlan 50,60,70,80,90,100
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface FastEthernet0/5
 switchport trunk allowed vlan 50,60,70,80,90,100
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active

interface FastEthernet0/6
 switchport trunk allowed vlan 100
 switchport trunk encapsulation dot1q
!
interface FastEthernet0/7
 switchport trunk allowed vlan 90
 switchport trunk encapsulation dot1q
!
interface FastEthernet0/8
 switchport trunk allowed vlan 80
 switchport trunk encapsulation dot1q
!
interface FastEthernet0/9
 switchport trunk allowed vlan 70
 switchport trunk encapsulation dot1q
!
interface FastEthernet0/10
 switchport trunk allowed vlan 60
!
interface FastEthernet0/11
 switchport trunk allowed vlan 50
 switchport trunk encapsulation dot1q
.

```

```

interface Vlan1
no ip address
shutdown
!
interface Vlan50
mac-address 0002.1626.4801
ip address 10.0.50.3 255.255.255.0
ip helper-address 10.0.50.2
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B5:A050::3/64
ipv6 nd other-config-flag
!
interface Vlan60
mac-address 0002.1626.4802
ip address 10.0.60.3 255.255.255.0
ip helper-address 10.0.60.2
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B6:A060::3/64
ipv6 nd other-config-flag
!
interface Vlan70
mac-address 0002.1626.4803
ip address 10.0.70.3 255.255.255.0
ip helper-address 10.0.70.2
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B7:A070::3/64
ipv6 nd other-config-flag
!
interface Vlan80
mac-address 0002.1626.4804
ip address 10.0.80.3 255.255.255.0
ip helper-address 10.0.80.2
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B8:A080::3/64
ipv6 nd other-config-flag

```

```

interface Vlan90
mac-address 0002.1626.4805
ip address 10.0.90.3 255.255.255.0
ip helper-address 10.0.90.2
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:B9:A090::3/64
ipv6 nd other-config-flag
!
interface Vlan100
mac-address 0002.1626.4806
ip address 10.0.100.3 255.255.255.0
ip helper-address 10.0.100.2
ipv6 nd ra suppress
ipv6 address 2001:0:CAFE:BA:A0A0::3/64
ipv6 nd other-config-flag
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
!
end

```

5. Configuration des switchs de type layer 2 :

a) Switch Serveur WEB :

```
Switch_serveur_web#show run
Building configuration...

Current configuration : 1200 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_serveur_web
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 30
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 30
 switchport mode access
```

b) Switch Réunion:

```
Switch_reunion#show run
Building configuration...

Current configuration : 1196 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_reunion
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 40
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 40
 switchport mode access
```

c) Switch Accueil:

```
Switch_accueil#show run
Building configuration...

Current configuration : 1196 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_accueil
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 10
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
```

d) Switch Showroom:

```
Switch_showroom#show run
Building configuration...

Current configuration : 1197 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_showroom
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 20
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
```

e) Switch Direction:

```
Switch_direction#show run
Building configuration...

Current configuration : 1258 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_direction
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 100
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 100
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 100
 switchport mode access
```

f) Switch RH:

```
Switch_RH#show run
Building configuration...

Current configuration : 1248 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_RH
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 90
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 90
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 90
 switchport mode access
```

g) Switch Commerciaux :

```
Switch_commercial#show run
Building configuration...

Current configuration : 1256 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_commercial
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 80
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 80
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 80
 switchport mode access
```

h) Switch Finance :

```
Switch_finance#show run
Building configuration...

Current configuration : 1253 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_finance
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 70
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 70
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 70
 switchport mode access
```

i) Switch Service Informatique :

```
Switch_informatique#show run
Building configuration...

Current configuration : 1292 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_informatique
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 60
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 60
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 60
 switchport trunk allowed vlan 60
 switchport mode access
```

j) Switch du serveur DNS :

```
Switch_serveur_DNS#show run
Building configuration...

Current configuration : 1257 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch_serveur_DNS
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 50
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk allowed vlan 50
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 50
 switchport mode access
!
```

